Lecture 14

Cramer's rule

Recall: Let A be an $n \times n$ matrix and $\mathbf{b}$ a column vector with n-entries, we shall denote by $A_k[\mathbf{b}]$ (where $1 \leq k \leq n$) the matrix obtained by replacing the k-th column of A by $\mathbf{b}$.

Example If $A = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 6 & 9 \\ \pi & 7 & 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix}$ then $A_2[\mathbf{b}] = \begin{pmatrix} 1 & 1 & 4 \\ 2 & 7 & 9 \\ \pi & 4 & 3 \end{pmatrix}$

Theorem 10. *(Cramer's rule) Let A be an invertible $n \times n$ matrix and $\boldsymbol{b} \in \mathbb{R}^n$, then the linear system $A\boldsymbol{x} = \boldsymbol{b}$ has a unique solution and is given by*

$$x_k = \frac{det(A_k[\boldsymbol{b}])}{det(A)}, \text{ for } k=1,2,\ldots, n.$$

Proof. We begin by checking that we have $AI_k[\mathbf{x}] = A_k[\mathbf{b}]$ (where $I_k[\mathbf{x}]$ is of cause the matrix obtained by replacing the k-th column of the idenity matrix by vector $\mathbf{x}$ of variables), the (i, j)-th entry of the LHS is given by $\displaystyle\sum_{l=1}^{n} a_{il}d_{lj}$ where d_{lj} is the (l, j)-th entry of $I_k[\mathbf{x}]$.
We consider two cases:
Case (1): $(k = j)$, in this case we have $d_{lk} = x_l$ and so the (i, j)-th entry of the LHS then becomes $\displaystyle\sum_{l=1}^{n} a_{il}x_l$ and since $A\mathbf{x} = \mathbf{b}$ we have that $\displaystyle\sum_{l=1}^{n} a_{il}x_l = b_i$ which is the (i, j)-th entry of RHS $A_k[\mathbf{b}]$ in this case.
Case (2): $(k \neq j)$, in this case we have that $d_{lj} = \begin{cases} 0 & \text{if } l \neq j \\ 1 & \text{if } l = j \end{cases}$ so the (i, j)-th entry of the LHS then becomes $\displaystyle\sum_{l=1}^{n} a_{il}d_{lj} = a_{ij}$ which is the (i, j)-th entry of the RHS $A_k[\mathbf{b}]$ in this case.

Now we have that $det(A)det(I_k[\mathbf{x}]) = det(A_k[\mathbf{b}])$, and since A is invertible we have $det(A) \neq 0$ and so $det(I_k[\mathbf{x}]) = \frac{det(A_k[\mathbf{b}])}{det(A)}$. We are now left to show that $det(I_k[\mathbf{x}]) = x_k$, we can expand this determinant across the k-th column of $I_k[\mathbf{x}]$, for each row $1 \leq i \leq n$, the associated minor $M_{ik} = \begin{cases} 0 & \text{if } i \neq k \\ 1 & \text{if } i = k \end{cases}$ (this is because for $i \neq k$, the i-th column of the matrix of the minor will consist only of zeros) and so $det(I_k[\mathbf{x}]) = \displaystyle\sum_{i=1}^{n} (-1)^{i+k} x_i M_{ik} = (-1)^{2k} x_k M_{kk} = x_k$. $\square$

Example Use Cramer's rule to find the solution of the linear system below:

$$\begin{aligned} 2x_1 - x_2 + 3x_3 &= 0 \\ 2x_2 + x_3 &= 0 \\ x_1 + x_3 &= 1 \end{aligned}$$

From the above linear system we have $A = \begin{pmatrix} 2 & -1 & 3 \\ 2 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
and so $A_1[\mathbf{b}] = \begin{pmatrix} 0 & -1 & 3 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$, $A_2[\mathbf{b}] = \begin{pmatrix} 2 & 0 & 3 \\ 2 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ and $A_3[\mathbf{b}] = \begin{pmatrix} 2 & -1 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

hence we have solution $x_1 = \frac{det(A_1[\mathbf{b}]}{det(A)} = \frac{-1}{1} = -1$, $x_2 = \frac{det(A_1[\mathbf{b}]}{det(A)} = \frac{4}{1} = 4$ and $x_3 = \frac{det(A_1[\mathbf{b}]}{det(A)} = \frac{2}{1} = 2$.

Vector spaces

Vector spaces can be defined in a number of equivalent ways and we shall adapt the definition in your notes:

Definition 28. *Given $F = \mathbb{R}$ or $\mathbb{C}$. Let V be a set, and $+ : V \times V \to V$ a binary operation on V taking pair $(\boldsymbol{v}, \boldsymbol{w}) \mapsto \boldsymbol{v} + \boldsymbol{w}$ and scalar multiplication $F \times V \to V$ taking pair $(\lambda, \boldsymbol{v}) \mapsto \lambda \boldsymbol{v}$. We call V (together with the above addition and scalar multiplication) a vector space (or linear space) over F if the following conditions hold for each $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in V$ and each $\lambda, \mu \in F$.*

1. $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$;
2. $\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$;
3. *There exists $\boldsymbol{0} \in V$ such that $\boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$ (for each $\boldsymbol{u} \in V$);*
4. *For each $\boldsymbol{u} \in V$, there exists $\boldsymbol{u}' \in V$ such that $\boldsymbol{u} + \boldsymbol{u}' = \boldsymbol{0}$;*
5. $\lambda(\boldsymbol{u} + \boldsymbol{v}) = \lambda \boldsymbol{u} + \lambda \boldsymbol{v}$;
6. $(\lambda + \mu)\boldsymbol{u} = \lambda \boldsymbol{u} + \mu \boldsymbol{u}$;
7. $(\lambda\mu)\boldsymbol{u} = \lambda(\mu \boldsymbol{u})$;
8. $1\boldsymbol{u} = \boldsymbol{u}$.

Remark 1. *In general, vector spaces can be defined over arbitrary fields F (a field is yet another mathematical structure) - do self study if interested. In this module we shall only look at vector spaces over the field of real numbers or complex numbers. The former are often called real vector spaces and the latter complex vector spaces.*

The $\mathbf{0}$ in axiom 3 is unique, for if we suppose we have two such $\mathbf{0}$ and $\mathbf{0'}$ we get $\mathbf{0'} = \mathbf{0'} + \mathbf{0} = \mathbf{0}$, similarly for each $\mathbf{u}$, the $\mathbf{u}'$ in axiom 4 is uniquely determined, again if we have two such $\mathbf{u}'$ and $\mathbf{u}''$, we get $\mathbf{u}' = \mathbf{u}' + \mathbf{0} = \mathbf{u}' + (\mathbf{u} + \mathbf{u}'') = (\mathbf{u}' + \mathbf{u}) + \mathbf{u}'' = \mathbf{0} + \mathbf{u}'' = \mathbf{u}''$, being uniquely determined, we shall denote this element $\mathbf{u}'$ by $-\mathbf{u}$.
Make sure you are able to carry out these little arguments independently.

Examples (1) Notice that axiom (3) requires existence of the $\mathbf{0}$ element in V, so we would consider the simplest vector space $V = \{\mathbf{0}\}$ over $F = \mathbb{R}$ or $\mathbb{C}$.
(2) Revisit the first lecture - the properties we were proving about $m \times n$ matrices precisely say that set $M_{m \times n}(\mathbb{C})$ or $M_{m \times n}(\mathbb{R})$ with component-wise addition and scalar multiplication satisfy the vector space axioms.
(3) $\mathbb{R}^n$ for each positive integer n form a real vector space under component-wise addition and scalar multiplication.
(4) Real-valued functions on a fixed domain $D \subseteq \mathbb{R}$ form a real vector space $\mathbb{R}^D$, define addition $f + g$ of such two functions f and g by $(f + g)(x) = f(x) + g(x)$ and scalar multiplication λf by $(\lambda f)(x) = \lambda f(x)$ for each $x \in D$. Vector space axioms are satisfied in this case since they are for each $x \in D$.

(4) Let P_n for each positive integer n denote the set of all polynomials in one variable with real coefficients and degree at most n. Given two such polynomials $p = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ and $q = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n$, define their addition and scalar multiplication coefficient-wise. (If you get a sense that this can be 'identified' with $\mathbb{R}^{n+1}$ you are not far from truth but we shall make precise what we mean by 'identify' later.)
(5) Think of many other examples of 'vector spaces' you have come across... Specify the set, addition and scalar multiplication and convince yourself that the vector space axioms are indeed satisfied.

Theorem 11. *Given $F = \mathbb{R}$ or $\mathbb{C}$. Let V be a vector space over F, if $\boldsymbol{u} \in V$ and $\lambda \in F$, then we have*

(a) $0\boldsymbol{u} = \boldsymbol{0}$*;*
(b) $\lambda\boldsymbol{0} = \boldsymbol{0}$*;*
(c) $(-1)\boldsymbol{u} = -\boldsymbol{u}$*;*
(d) If $\lambda\boldsymbol{u} = \boldsymbol{0}$*, then either* $\lambda = 0$ *or* $\boldsymbol{u} = \boldsymbol{0}$*.*

[Try to prove the above independently at least once]

Proof. (a) Using axiom (6) we have $(0+0)\mathbf{u} = 0\mathbf{u} + 0\mathbf{u}$, that is $0\mathbf{u} = 0\mathbf{u} + 0\mathbf{u}$ but since $0\mathbf{u}$ is an element in V, by axiom 4, there exists element $-(0\mathbf{u})$ such that $0\mathbf{u} + (-(0\mathbf{u})) = \mathbf{0}$, and so adding such an element to both sides of equation $0\mathbf{u} = 0\mathbf{u} + 0\mathbf{u}$ we get $0\mathbf{u} = \mathbf{0}$.
(b) Using axiom (5) we have $\lambda(\mathbf{0}+\mathbf{0}) = \lambda\mathbf{0} + \lambda\mathbf{0}$, that is $\lambda\mathbf{0} = \lambda\mathbf{0} + \lambda\mathbf{0}$, from here use similar argument as in (a) to deduce that $\lambda\mathbf{0} = \mathbf{0}$
(c) Using axiom (6), we have $(1 + (-1))\mathbf{u} = 1\mathbf{u} + (-1)\mathbf{u}$, that is, $0\mathbf{u} = 1\mathbf{u} + (-1)\mathbf{u}$, and from (a) and axiom (8) we get $\mathbf{0} = \mathbf{u} + (-1)\mathbf{u}$ and so $-\mathbf{u} = (-1)\mathbf{u}$ using axiom (4).
(d) Suppose $\lambda\mathbf{u} = \mathbf{0}$ and $\lambda \neq 0$, then we get $\lambda^{-1}(\lambda\mathbf{u}) = \lambda^{-1}\mathbf{0}$ and so we have $(\lambda^{-1}\lambda)\mathbf{u} = \mathbf{0}$ by axiom (7) and (b) and finally $\mathbf{u} = 1\mathbf{u} = 0$. $\square$